# Python Programming
# Unit 06 – Lecture 06 Notes
# Plot Types and Subplots

Tofik Ali

February 20, 2026

## Contents



## 1 Lecture Overview

Different plot types are used for different questions:

- trends → line plots,
- comparisons → bar charts,
- distributions → histograms,
- relationships → scatter plots.

Subplots help you combine multiple plots in one figure.

# 2 Core Concepts

## 2.1 Line Plot

Good for trends (time series).

```
plt.plot(x, y)
```

## 2.2 Bar Chart

Good for category comparisons.

```
plt.bar(categories, values)
```

## 2.3 Histogram

Good for distributions.

```
plt.hist(values, bins=10)
```

## 2.4 Scatter Plot

Good for relationships between two numeric variables.

```
plt.scatter(x, y)
```

## 2.5 Pie Chart (Use Carefully)

Pie charts show proportions, but can be hard to compare precisely.

```
plt.pie(values, labels=labels, autopct="%1.1f%%")
```

## 2.6 Subplots

Create multiple axes in one figure:

```
fig, ax = plt.subplots(2, 2, figsize=(8, 6))
ax[0, 0].plot(x, y)
```

# 3 Demo Walkthrough

**File:** `demo/matplotlib_plot_types_demo.py`
The demo creates a 2x2 subplot figure and saves it into `images/`.

# 4 Interactive Checkpoints (with Solutions)

## Checkpoint 1 Solution

**Question:** distribution of marks?
**Answer:** Histogram.

### Checkpoint 2 Solution

**Question:** compare marks of 5 subjects?
**Answer:** Bar chart.

## 5 Practice Exercises (with Solutions)

### Exercise 1: Histogram of Random Numbers

**Solution:**

```
import numpy as np
import matplotlib.pyplot as plt

x = np.random.randn(1000)
plt.hist(x, bins=20)
plt.title("Histogram")
plt.show()
```

## 6 Exit Question (with Solution)

**Question:** function name used to create subplots?
**Answer:** `subplots`

## Appendix: Slide Deck Content (Reference)

The material below is a reference copy of the slide deck content. Exercise solutions are explained in the main notes where applicable.

### Title Slide

Repository: `https://github.com/tali7c/Python-Programming`

## Quick Links

**Core Concepts** **Demo** **Interactive** **Summary**

## Agenda

- Core Concepts
- Demo
- Interactive
- Summary

## Learning Outcomes

- Create different plot types: line, bar, histogram, scatter, pie
- Build multi-plot figures using subplots
- Choose the right plot type for the data story

## Common Plot Types

- Line plot: trend over time
- Bar chart: compare categories
- Histogram: distribution of values
- Scatter plot: relationship between two variables
- Pie chart: parts of a whole (use carefully)

## Subplots

- Subplots allow multiple plots in one figure
- Use `plt.subplots(rows, cols)`

```
fig, ax = plt.subplots(2, 2)
ax[0, 0].plot(x, y)
```

## Demo: Multiple Plot Types in One Figure

- File: `demo/matplotlib_plot_types_demo.py`
- Generates a figure with subplots and saves it to `images/`

## Checkpoint 1

**Question:** Which plot type would you use to show distribution of marks of 100 students?

## Checkpoint 2

**Question:** Which plot type would you use to compare marks of 5 subjects?

## Think-Pair-Share

Discuss:

- Why can pie charts be misleading?

## Key Takeaways

- Different plot types answer different questions
- Subplots help compare multiple views of data in one figure
- Always label and choose scales carefully for honest visuals

## Exit Question

Write the function name used to create subplots in Matplotlib.